ENERGY:

$$\frac{\partial}{\partial t}\left[\int_{Vg(t)} \rho\left(e+\frac{|\bar v|^2}{2}\right)\partial V\right] = \dot W_{est} + \dot Q$$

- conduction through the Σf
- volumetric source/sink → radiation, chemical reaction

$$\dot W_{est} = -\int_{\Sigma g(t)} p\bar n\cdot\bar v\,\partial\sigma + \int_{\Sigma g(t)} (\bar{\bar\tau}'\cdot\bar n)\cdot\bar v\,\partial\sigma + \int_{Vg(t)} \rho\bar f_m\cdot\bar v\,\partial V$$

$$-\int_{\Sigma g(t)} q_n(\bar r,\bar x,t)\,\partial\sigma = -\int_{\Sigma g(t)} \bar q\cdot\bar n\,\partial\sigma = +\int_{\Sigma g(t)} k\nabla T\cdot\bar n\,\partial\sigma$$

q_n: heat flux p.u. surface transfered to the fluid element located where $\bar n$ points to

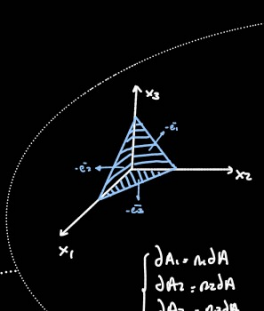
How q_n depends on $\bar n$? → $q_n = \bar q\cdot\bar n$

$\bar q = -k\nabla T$ (Fourier's Law)

$$\dot Q = -\int_{\Sigma g(t)} \bar q\cdot\bar n\,\partial\sigma + \int_{Vg(t)} \dot Q_{rc}\,\partial V$$

(Heat added) p.u. time to the Vg

$\dot Q_{rc}$: heat p.u. time p.u. volume added to the fluid by radiation or chemical reaction



$q_n\,\partial A$: heat p.u. time through surface ∂A, $\bar n$.
$q_1\,\partial A_1$: " " " " " ∂A_1, $\bar e_1$.
$q_2\,\partial A_2$: " " " " " ∂A_2, $\bar e_2$.
$q_3\,\partial A_3$: " " " " " ∂A_3, $\bar e_3$.

$\partial A_1 = n_1\partial A$, $\partial A_2 = n_2\partial A$, $\partial A_3 = n_3\partial A$

$q_n\,\partial A - q_1\partial A_1 - q_2\partial A_2 - q_3\partial A_3 = 0$

$q_n - q_1 n_1 - q_2 n_2 - q_3 n_3 = 0$

$q_n = q_1 n_1 + q_2 n_2 + q_3 n_3 \rightarrow \bar q\cdot\bar n$

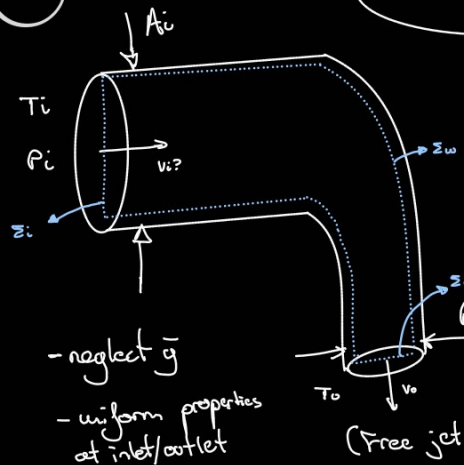
$$\frac{\partial}{\partial t}\left[\int_{Vg(t)} \rho\left(e+\frac{|\bar v|^2}{2}\right)\partial V\right] = -\int_{\Sigma g} p\bar v\cdot\bar n\,\partial\sigma + \int_{\Sigma g(t)} (\bar{\bar\tau}'\cdot\bar n)\cdot\bar v\,\partial\sigma + \int_{Vg(t)} \rho\bar f_m\cdot\bar v\,\partial V - \int_{\Sigma g(t)} \bar q\cdot\bar n\,\partial\sigma + \int_{Vg(t)} \dot Q_{rc}\,\partial V$$

RTT

$$\frac{\partial}{\partial t}\left[\int_{Vc(t)} \rho\left(e+\frac{|\bar v|^2}{2}\right)\partial V\right] + \int_{\Sigma c(t)} \rho\left(e+\frac{|\bar v|^2}{2}\right)(\bar v-\bar v_c)\cdot\bar n\,\partial\sigma = \text{" " " " " "}$$

CL6

a) v_i=? b) $|F_x, F_z|$=? c) $\dot Q$=? (From pipe wall to liq)



a) mass: $\int_{\Sigma c(t)} \rho(\bar v - \bar v_c)\cdot\bar n\,\partial\sigma = 0$ ($\bar v_c = 0$)

Σ_i	$\rho v_i\bar e_x\cdot(-\bar e_x)A_o = -\rho v_i A_i$
Σ_o	$\rho(-v_o\bar e_y)\cdot(-\bar e_y)A_o = +\rho v_o A_o$
Σ_ω	0

$v_i A_i = v_o A_o \rightarrow \boxed{v_i = \frac{v_o A_o}{A_i}}$

b) $\bar F_{liq\to pipe} = -\int_{\Sigma_\omega} p\bar n_{liq}\,\partial\sigma + \int_{\Sigma_\omega} \bar{\bar\tau}'\cdot\bar n_{liq}\,\partial\sigma - \int_{\Sigma_\omega} p_a\bar n_{air}\,\partial\sigma = -\int_{\Sigma_\omega} (p-p_a)\bar n_{liq}\,\partial\sigma + \int_{\Sigma_\omega} \bar{\bar\tau}'\cdot\bar n_{liq}\,\partial\sigma$ (?)

$$\Rightarrow \boxed{\bar F = \rho v_i^2 A_i\bar e_x + \rho v_o^2 A_o\bar e_y + (p_i-p_a)A_i\bar e_x = \dot m(v_i\bar e_x + v_o\bar e_y) + (p_i-p_a)A_i\bar e_x}$$

$\dot m = \rho v_o A_o = \rho v_i A_i$

momentum: $\frac{\partial}{\partial t}\int_{Vc(t)} \rho\bar v\,\partial V + \int_{\Sigma c(t)} \rho\bar v(\bar v-\bar v_c)\cdot\bar n\,\partial\sigma = -\int_{\Sigma c} (p-p_a)\bar n\,\partial\sigma + \int_{\Sigma c} \bar{\bar\tau}'\cdot\bar n\,\partial\sigma + \int_{Vc} \rho\bar g\,\partial V$ (neglect $\bar g$)

Σ_i	$-\rho v_i^2 A_i\bar e_x$	Σ_i	$(p_i-p_a)A_i\bar e_x$
Σ_o	$-\rho v_o^2 A_o\bar e_y$	Σ_o	0
Σ_ω	0	Σ_ω	$-\int_{\Sigma_\omega}(p-p_a)\bar n\,\partial\sigma$
		Σ_ω	$\int_{\Sigma_\omega}\bar{\bar\tau}'\cdot\bar n\,\partial\sigma$

$\rightarrow -\bar F$

c) energy: $\int_{\Sigma_c}\left(e+\frac{|\bar v|^2}{2}\right)(\bar v-\bar v_c)\cdot\bar n\,\partial\sigma = -\int_{\Sigma_c} p\bar n\cdot\bar v\,\partial\sigma + \int_{\Sigma_c}\bar{\bar\tau}'\cdot\bar n\cdot\bar v\,\partial\sigma + \int_{Vc}\rho\bar g\cdot\bar v\,\partial V - \int_{\Sigma_c}\bar q\cdot\bar n\,\partial\sigma + \int_{Vc}\dot Q_{rc}\,\partial V$ (neglect $\bar g$)

$$\Rightarrow \boxed{\dot Q = \rho v_o A_o\left[c(T_o-T_i) + \frac{v_o^2-v_i^2}{2}\right] + (p_a-p_i)v_o A_o}$$

Σ_i	$\left(cT_i + \frac{v_i^2}{2}\right)(-v_i)\rho A_i$	Σ_i	$p_i v_i A_i$
Σ_o	$\left(cT_o + \frac{v_o^2}{2}\right)\rho v_o A_o$	Σ_o	$-p_a v_o A_o$
Σ_ω	0	Σ_ω	0
		Σ_ω	$-\int_{\Sigma_\omega} p\bar v\cdot\bar n\,\partial\sigma$
		Σ_ω	$\int_{\Sigma_\omega}\bar{\bar\tau}'\cdot\bar n\cdot\bar v\,\partial\sigma$
		Σ_ω	$-\int_{\Sigma_\omega}\bar q\cdot\bar n\,\partial\sigma$

Σ_{in} 0

Σ_o $-\rho_o v_o A_o$

Σ_w 0

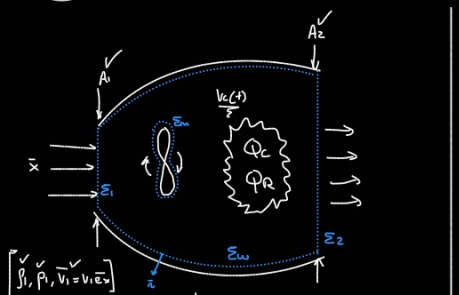
Σ_o 0

Σ_w 0

Σ_o 0

Σ_w $-\int \bar{q}\cdot\bar{n}\,d\sigma \rightarrow \dot{Q}$

CL5



- Compressible
- Neglect gravity
- Uniform prop. at inlet/outlet

$\dot{W}$: Work p.u. time extracted from fluid

$\dot{Q}$: heat transferred p.u. time to the fluid

$\bar{F}$: force on the device, measured on test bench

$[\bar{F} = -F\bar{e}_{ax}]$

$\bar{F}_{fluid\ inside\ amb\ air} \rightarrow d\bar{e}v = -\int_{\Sigma_m,\Sigma_w} p\bar{n}_{inside}\,d\sigma + \int_{\Sigma_m,\Sigma_w} \bar{\bar{\tau}}'\cdot\bar{n}_{inside}\,d\sigma - \int_{\Sigma_w} p_a \underbrace{\bar{n}_{amb}}_{(-\bar{n}_{inside})}\,d\sigma = \int_{\Sigma_m,\Sigma_w} (p-p_a)\bar{n}_{inside}\,d\sigma + \int_{\Sigma_m,\Sigma_w} \bar{\bar{\tau}}'\cdot\bar{n}_{inside}\,d\sigma$

$\rightarrow \dot{Q} = \int_{\Sigma_w,\Sigma_m} \bar{q}\cdot\bar{n}_{inside}\,d\sigma + \int_{V_c} Q_R\,dV$ ($\bar{n}_{inside}$: pointing to the fluid)

$\rightarrow \dot{W}_{fluid-device} = -\int_{\Sigma_m} p\bar{n}_{inside}\cdot\bar{v}\,d\sigma + \int_{\Sigma_m} \bar{\bar{\tau}}'\cdot\bar{n}_{inside}\cdot\bar{v}\,d\sigma$ (only Σ_m as Σ_w is a fixed surface (0 work on it))

mass: $\frac{\partial}{\partial t}\int_{V_c(t)} \rho\,dV + \int_{\Sigma_c(t)} \rho(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma = 0 \Rightarrow \rho_1 v_1 A_1 = \rho_2 v_2 A_2$ (1)

$\dot{m}_1 = \dot{m}_2 = \dot{m}$

Σ_1	$-\rho_1 v_1 A_1$
Σ_2	$\rho_2 v_2 A_2$
Σ_m	$(\bar{v}=\bar{v}_c \neq 0) \Rightarrow 0$
Σ_w	0

momentum: $\frac{\partial}{\partial t}\int_{V_c(t)} \rho\bar{v}\,dV + \int_{\Sigma_c(t)} \rho\bar{v}(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma = -\int_{\Sigma_c(t)} (p-p_a)\bar{n}\,d\sigma + \int_{\Sigma_c(t)} \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \int_{V_c(t)} \rho\bar{g}\,dV$ (neglect) $\rightarrow \dot{m}(v_c - v_1) = (p_1 - p_a)A_1 - (p_2 - p_a)A_2 + F$ (2)

Σ_1	$\rho_1 v_1 \bar{e}_{1x}(-v_1)A_1 = -\dot{m}v_1\bar{e}_{1x}$	Σ_1	$-(p_1-p_a)(-\bar{e}_{1x})A_1 = (p_1-p_a)A_1\bar{e}_{1x}$
Σ_2	$\rho_2 v_2 \bar{e}_{2x}(v_2)A_2 = \dot{m}v_2\bar{e}_{2x}$	Σ_2	$-(p_2-p_a)A_2\bar{e}_{2x}$
Σ_m	0	Σ_m	$\int_{\Sigma_m,\Sigma_w} -(p-p_a)\bar{n}\,d\sigma + \int_{\Sigma_m,\Sigma_w} \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma \rightarrow -\bar{F} = F\bar{e}_{ax}$
Σ_w	0	Σ_w	

Σ_1, Σ_2 0 (uniform velocity vectors)

energy: $\int_{\Sigma_c(t)} \rho\left(e + \frac{|\bar{v}|^2}{2}\right)(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma = -\int_{\Sigma_c(t)} p\bar{n}\cdot\bar{v}\,d\sigma + \int_{\Sigma_c(t)} (\bar{\bar{\tau}}'\cdot\bar{n})\cdot\bar{v}\,d\sigma - \int_{\Sigma_c(t)} \bar{q}\cdot\bar{n}\,d\sigma + \int_{V_c(t)} Q_R\,dV \Rightarrow \dot{m}\left[\left(e_2 + \frac{v_2^2}{2}\right) - \left(e_1 + \frac{v_1^2}{2}\right)\right] = \int_{\Sigma_1}\rho v_1 A_1 - \int_{\Sigma_2}\rho v_2 A_2 - \dot{W} + \dot{Q}$ ($\bar{q} = -k\nabla T$)

Σ_1	$-\rho_1\left(e_1 + \frac{v_1^2}{2}\right)v_1 A_1 = -\dot{m}\left(e_1 + \frac{v_1^2}{2}\right)$	Σ_1	$+p_1 v_1 A_1$
Σ_2	$\rho_2\left(e_2 + \frac{v_2^2}{2}\right)v_2 A_2 = \dot{m}\left(e_2 + \frac{v_2^2}{2}\right)$	Σ_2	$-p_2 v_2 A_2$
$\Sigma_m(t)$	$-\int \rho\bar{v}\cdot\bar{n}\,d\sigma \rightarrow -\dot{W}$	Σ_m	$\int_{\Sigma_m} \bar{\bar{\tau}}'\cdot\bar{v}\,d\sigma$
Σ_w	0	Σ_w	0

Σ_1, Σ_2 0 ($\nabla T = 0$)

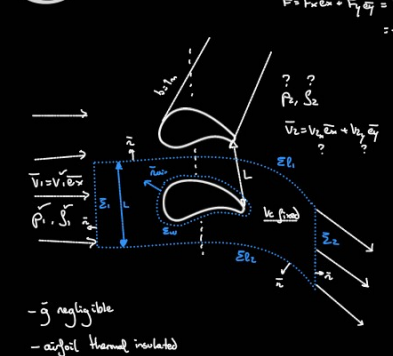
Σ_m, Σ_w $-\int_{\Sigma_m,\Sigma_w} \bar{q}\cdot\bar{n}\,d\sigma \rightarrow \dot{Q}$

$\Rightarrow \dot{m}\left[\left(h_2 + \frac{v_2^2}{2}\right) - \left(h_1 + \frac{v_1^2}{2}\right)\right] = -\dot{W} + \dot{Q}$ (3)

$h_2 = u_2 + \frac{p_2}{\rho_2} = c_v T_2 + \frac{p_2}{\rho_2} = c_v\frac{p_2}{\rho_2 R} + \frac{p_2}{\rho_2} = \frac{c_v + R}{R}\frac{p_2}{\rho_2} = \frac{\gamma}{\gamma-1}\frac{p_2}{\rho_2}$ ($\rho = \rho_0 \frac{p}{p_0}$, $\gamma = \frac{c_p}{c_v}$)

$h_1 = \frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1}$

CL4



$\bar{F} = F_x\bar{e}_{1x} + F_y\bar{e}_{2y} = -F_{ax}$ (on airfoil (on body)) $= -\int_{\Sigma_w} p\bar{n}\,d\sigma + \int_{\Sigma_w} \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

mass: $\int_{\Sigma_c} \rho(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma = 0 \Rightarrow \rho_1 v_1 = \rho_2 v_{2x}$ (1)

Σ_1	$\rho_1(-v_1)L\,b$
Σ_2	$\rho_2 v_{2x} L\,b$

momentum: $\int_{\Sigma_2} \rho\bar{v}(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma = -\int_{\Sigma_c} p\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$ (confirm operations ?)

$\Rightarrow \begin{cases} F_x = \rho_1 v_1 L(v_1 - v_{2x}) + (p_1 - p_2)L & (2) \\ F_y = -\rho_1 v_1 L\,v_{2y} & (3) \end{cases}$

Σ_1	$\rho_1 v_1 \bar{e}_{1x}(-v_1)Lb$	Σ_1	$+p_1\bar{e}_{1x}Lb$
Σ_2	$\rho_2(v_{2x}\bar{e}_{1x} + v_{2y}\bar{e}_{2y})v_{2x}Lb$	Σ_2	$-p_2\bar{e}_{1x}Lb$
		Σl_1	$-\int_{\Sigma l_1} p\bar{n}\,d\sigma$?
		Σl_2	$-\int_{\Sigma l_2} p\bar{n}\,d\sigma$? $\Rightarrow \Sigma l_1 + \Sigma l_2 = 0$
		Σ_w	$-\int_{\Sigma_w} p\bar{n}\,d\sigma$

Σ_1	0
Σ_2	0
Σl_1	$\Sigma l_1 + \Sigma l_2 = 0$
Σl_2	
Σ_w	$\int_{\Sigma_w} \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma \rightarrow -\bar{F} = -F_x\bar{e}_{1x} - F_y\bar{e}_{2y}$

energy: $\frac{\partial}{\partial t}\int_{V_c} \rho\left(e + \frac{|\bar{v}|^2}{2}\right)dV + \int_{\Sigma_c(t)} \rho\left(e + \frac{|\bar{v}|^2}{2}\right)(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma = -\int_{\Sigma_c(t)} p\bar{n}\cdot\bar{v}\,d\sigma + \int_{\Sigma_c(t)} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \int_{V_c(t)} \rho\bar{g}\cdot\bar{v}\,dV$ (neglect) $- \int_{\Sigma_c(t)} \bar{q}\cdot\bar{n}\,d\sigma + \int_{V_c} Q_R\,dV \rightarrow \rho_1 v_1 L\left(e_2 + \frac{v_{2x}^2 + v_{2y}^2}{2} - \left(e_1 + \frac{v_1^2}{2}\right)\right) = \left(p_1\frac{\rho_1}{\rho_1}v_1 L - p_2\frac{\rho_2}{\rho_2}v_{2x}L\right)$

$\Rightarrow \left[\left(e_2 + \frac{p_2}{\rho_2}\right) + \frac{v_{2x}^2 + v_{2y}^2}{2} - \left(e_1 + \frac{p_1}{\rho_1} + \frac{v_1^2}{2}\right)\right] = 0$

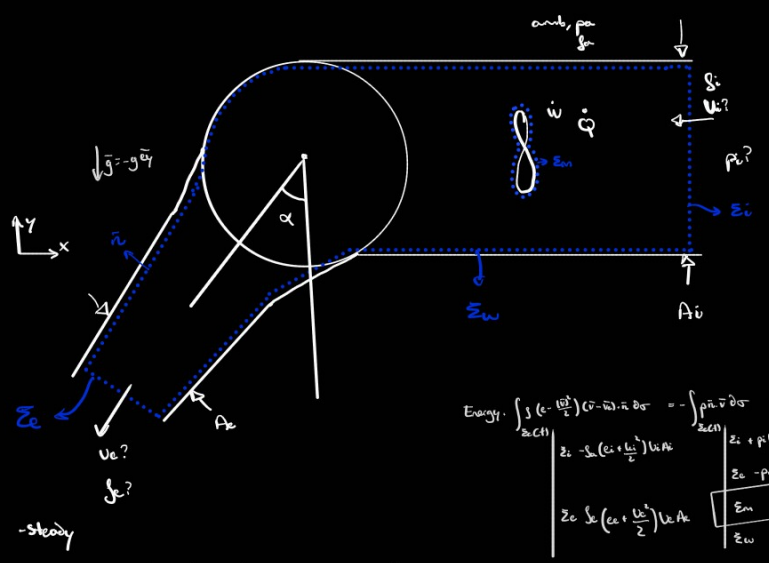
$h_2 \qquad\qquad h_1$

$\Rightarrow \frac{\gamma}{\gamma-1}\left(\frac{p_2}{\rho_2} - \frac{p_1}{\rho_1}\right) + \frac{v_{2x}^2 + v_{2y}^2}{2} - \frac{v_1^2}{2} = 0$ (4)

Σ_1	$\rho_1\left(e_1 - \frac{v_1^2}{2}\right)(-v_1)L$	Σ_1	$+p_1 v_1 L$
Σ_2	$\rho_2\left(e_2 + \frac{v_{2x}^2 + v_{2y}^2}{2}\right)v_{2x}L$	Σ_2	$-p_2 v_{2x} L$
Σl_1	0	Σl_1	
Σl_2	0	Σl_2	0 (cancel each other)
Σ_w	0	Σ_w	

Σ_1	0
Σ_2	0
Σl_1	0
Σl_2	
Σ_w	0

CL 11



Mass. $\dot{m}_a = \dot{m}_i = \dot{m}_e$ ⇒ $\rho_i U_i A_i = \rho_e U_e A_e$ ①

Mom. $-\dot{m}_i U_i \bar{e}_{Ai} - \dot{m}_e U_e (\cos\alpha\, \bar{e}_x + \sin\alpha\, \bar{e}_y) = -(p_i - p_a) A_i \bar{e}_x - \bar{F}$

x: $-\dot{m}_e U_i - \dot{m}_e U_e \sin\alpha = -(p_i - p_a)A_i$ ②

y: $+\dot{m}_e U_e \cos\alpha = +Mg$ ③

Energy. $\int_{\Sigma_c(t)} \left(e + \frac{U^2}{2}\right)(\vec{v}-\vec{w})\cdot\vec{n}\,\partial\sigma = -\int_{\Sigma_c(t)} p\vec{v}\cdot\vec{n}\,\partial\sigma + \int_{\Sigma_c(t)} \vec{v}\cdot(\vec{\tau}\cdot\vec{n})\,\partial\sigma - \int_{\Sigma_c(t)} \vec{q}\cdot\vec{n}\,\partial\sigma + \int_{V_c} \rho r_e\,\partial V$

$\Sigma_c(t)$	Σ_i	Σ_a	Σ_e	Σ_m	Σ_w
$-\dot{m}_i\left(e_i + \frac{U_i^2}{2}\right)$	$\dot{m}_i\left(e_i + \frac{U_i^2}{2}\right)$	$p_a U_i A_i$	$-p_a U_e A_e$	$\dot{W}$	0
$\dot{m}_e\left(e_e + \frac{U_e^2}{2}\right)$	0	0	0	0	$\dot{Q}$

⇓

$$-\dot{m}_i\left(e_i + \frac{U_i^2}{2}\right) + \dot{m}_e\left(e_e + \frac{U_e^2}{2}\right) = \frac{\dot{m}_a}{\rho_a}\rho_i U_i A_i - \frac{\dot{m}_e}{\rho_e} p_a U_e A_e + \dot{W} + \dot{Q}$$

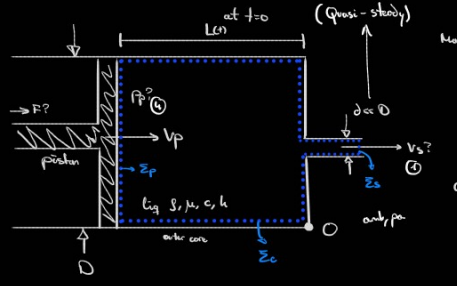
$$\dot{m}\left[\underbrace{e_e + \frac{p_a}{\rho_e}}_{h_e = \frac{\gamma}{\gamma-1}\frac{p_a}{\rho_e}} + \frac{U_e^2}{2} - \left(\underbrace{e_i + \frac{p_i}{\rho_a}}_{h_i = \frac{\gamma}{\gamma-1}\frac{p_i}{\rho_a}} + \frac{U_i^2}{2}\right)\right] = \dot{W} + \dot{Q}$$ ④

- steady

- $\dot{W}$: work p.u. time done by device ⇒ air stream
- $\dot{Q}$: heat added p.u. time to the air stream
- $\bar{F}$: force { air stream, amb. air } → device $F_x = 0$, $F_y = Mg$

- Bernoulli: $p_a = p_i + \frac{1}{2}\rho_a U_i^2$ ⑤ (Total pressure)

Extra Exercise



Mass. $\frac{\partial}{\partial t}\int_{V_c(t)} \rho\,\partial V + \int_{\Sigma_c(t)} \rho(\vec{v}-\vec{w})\cdot\vec{n}\,\partial\sigma = 0 \Rightarrow \rho_p D^2 = \rho_s \delta^2 \Rightarrow \boxed{V_s = V_p\left(\frac{D}{\delta}\right)^2}$ ①

$\int \frac{\partial \rho_a}{\partial t}$ — Cross-section of outlet orifice $-\rho_p$ — $\frac{\partial L}{\partial t}$ — $-\rho\frac{\pi D^2}{4}\rho_p$

$\Sigma_c(t)$	Σ_p	Σ_c	Σ_s
	$\vec{v}=0$; $\vec{v}_s\cdot\vec{n} = \rho_p\vec{e}_x$	0	
	ρ_p	0	$\rho V_s \pi\frac{\delta^2}{4}$

$\vec{v} = V_p\vec{e}_x$

$$\frac{\partial}{\partial t}\int_{V_c(t)} \rho\vec{v}\,\partial V + \int_{\Sigma_c(t)} \rho\vec{v}(\vec{v}-\vec{w})\cdot\vec{n}\,\partial\sigma = -\int_{\Sigma_c(t)} (p-p_a)\vec{n}\,\partial\sigma + \int_{\Sigma_c(t)} \vec{\tau}\cdot\vec{n}\,\partial\sigma + \int_{V_c(t)} \rho\vec{g}\,\partial V$$ (× neglect)

$\Sigma_c(t)$	Σ_p	Σ_c	Σ_s
	ρ_p	0	$\rho V_p^2\frac{\pi D^2}{4}\rho_p$
	$\bar{F}_p$	$\bar{F}_c$	

$\bar{F}_p + \bar{F}_c = -\rho V_s^2\frac{\pi\delta^2}{4}\vec{e}_x$ ②

Unsteady / Convective $\sim \frac{\rho V_p \frac{\pi D^2}{4}}{\rho V_s^2 \frac{\pi\delta^2}{4}}$ — $\frac{V_p^2 D^2}{V_p^2\left(\frac{D}{\delta}\right)^4\delta^2} = \frac{\delta^2}{D^2} \ll 1$ — we neglect that term (Quasi-steady)

② $(\bar{F}_p + \bar{F}_c)$? fluid inside / amb. air } injector

③ $(\bar{M}_{p,0} + \bar{M}_{c,0})$?

④ P_p?
- neglect $\vec{\tau}\cdot\vec{n}$
- quasi-steady
- incompr.

$P_p + \frac{1}{2}\rho V_p^2 = p_a + \frac{1}{2}\rho V_s^2 \Rightarrow P_p - p_a = \frac{1}{2}\rho(V_s^2 - V_p^2) \Rightarrow \boxed{P_p = p_a + \frac{1}{2}\rho(V_s^2 - V_p^2)}$ ⑥ → Finish ③ & ⑤

⑤ $\bar{F}_p$? $\bar{F}_c$?

4. Navier-Stokes Equation

- Mass conservation: $\frac{\partial}{\partial t}\int_{V_g(t)} \rho\,\partial V = 0$ (2.6) → $\int_{V_g(t)} \frac{\partial \rho}{\partial t}\partial V + \int_{\Sigma_g(t)} \rho\vec{v}\cdot\vec{n}\,\partial\sigma = 0$ (2.8) → $\int_{V_g(t)} \frac{\partial \rho}{\partial t}\partial V + \int_{V_g(t)} \nabla\cdot(\rho\vec{v})\,\partial V = 0$ (2.9)

- Leibniz's Rule

(3.6) $\frac{\partial}{\partial t}\int_{V_g(t)} \phi(\vec{x},t)\,\partial V = \int_{V_g(t)} \frac{\partial \phi}{\partial t}\partial V + \int_{\Sigma_g(t)} \phi\,\vec{v}\cdot\vec{n}\,\partial\sigma$

- Gauss Theorem

(2.8) $\oint_\Sigma \phi\vec{n}\,\partial\sigma = \int \nabla\phi\,\partial V$

$\int_{V_g(t)} \left[\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\vec{v})\right]\partial V = 0$ (0) → $\boxed{\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\vec{v}) = 0}$ 1st Navier-Stokes Eq. (Continuity)

$\frac{\partial \rho}{\partial t} + (\vec{v}\cdot\nabla)\rho + \rho\nabla\cdot\vec{v} = 0 \rightarrow \frac{D\rho}{Dt} + \rho\nabla\cdot\vec{v} = 0 \Rightarrow \frac{1}{\rho}\frac{D\rho}{Dt} = -\nabla\cdot\vec{v}$ (∵ ρ = constant) → $\nabla\cdot\vec{v} = 0$

thickness product $\int dh$
→ scalar or vectorial

$\frac{D\beta}{Dt}$

- Momentum Conservation : $\frac{\partial}{\partial t}\int_{V_{fg}(t)} \rho\vec{v}\,\partial V = \int_{V_{fg}(t)} \frac{\partial \rho\vec{v}}{\partial t}\partial V + \int_{\Sigma_{fg}(t)} \rho\vec{v}\,\underbrace{(\vec{v}\cdot\vec{n})}_{(\rho\vec{v})\cdot\vec{n}}\partial\Sigma = -\int_{\Sigma_{fg}(t)} p\vec{n}\,\partial\Sigma + \int_{\Sigma_{fg}(t)} \vec{\bar{\tau}}\cdot\vec{n}\,\partial\Sigma + \int_{V_{fg}(t)} \rho\vec{f}_m\,\partial V$

$$\left| \frac{\partial(\rho\vec{v})}{\partial t} + \nabla\cdot(\rho\vec{v}\vec{v}) = -\nabla p + \nabla\cdot\vec{\bar{\tau}} + \rho\vec{f}_m \right| \partial V$$

$$\vec{v}\frac{\partial\rho}{\partial t} + \rho\frac{\partial\vec{v}}{\partial t} + \vec{v}\nabla\cdot(\rho\vec{v}) + (\rho\vec{v}\cdot\nabla)\vec{v} \Rightarrow \boxed{\rho\frac{D\vec{v}}{Dt} = -\nabla p + \nabla\vec{\bar{\tau}} + \rho\vec{f}_m}$$

0 ; $\rho\frac{D\vec{v}}{Dt}$

(2nd Navier-Stokes Eq. (Newton's 2nd law for fluids))

$$\rho\frac{\partial\vec{v}}{\partial t} + \rho\vec{v}\cdot\nabla\vec{v} = \rho\frac{\partial\vec{v}}{\partial t} + \rho\nabla\frac{|\vec{v}|^2}{2} - \rho\vec{v}\wedge(\underbrace{\nabla\wedge\vec{v}}_{\vec{\omega}})$$

Bernoulli's eq.:
- Steady flow
- Incompressible (constant ρ)
- Negligible viscous forces ($\nabla\cdot\vec{\bar{\tau}} = 0$)
- Conservative mass forces $\vec{f}_m = -\nabla U$ ($-g\vec{e}_z = -(-g\vec{e}_z)\cdot\vec{z} = gz$)

$\vec{f}_m = \vec{g} \rightarrow U = -\vec{g}\cdot\vec{x} = gz$

$$\rho\nabla\left(\frac{|\vec{v}|^2}{2}\right) - \rho\vec{v}\wedge(\nabla\wedge\vec{v}) = -\nabla p - \rho\nabla U = -\nabla\underbrace{(p+\rho U)}_{P}$$

$$\left| \nabla\left(\rho\frac{|\vec{v}|^2}{2} + p + \rho U\right) = \rho\vec{v}\wedge(\nabla\wedge\vec{v}) \right| \cdot \frac{\vec{v}}{|\vec{v}|}$$

- for $\vec{\omega} = 0$, $\frac{1}{2}|\vec{v}|^2 + p + \rho U = \text{Constant}$
- for $\vec{\omega} \neq 0$, $\frac{\partial}{\partial s}\left(\rho\frac{|\vec{v}|^2}{2} + p + \rho U\right) = 0 \Rightarrow \boxed{\rho\frac{|\vec{v}|^2}{2} + p + \rho U = C_{ste}}$ Bernoulli's Eq

(Direction along the streamline) ; (Constant along a streamline direction)

- Energy: $\frac{\partial}{\partial t}\int_{V_{fg}(t)} \rho\left(e + \frac{|\vec{v}|^2}{2}\right)\partial V = \int_{V_{fg}(t)} \frac{D}{\partial t}\left(e + \frac{|\vec{v}|^2}{2}\right)\partial V + \int_{\Sigma_{fg}(t)} \rho\left(e + \frac{|\vec{v}|^2}{2}\right)\vec{v}\cdot\vec{n}\,\partial\Sigma$ (Leibniz's Rule)

$$= -\int_{\Sigma_{fg}} p\vec{v}\cdot\vec{n}\,\partial\Sigma + \int_{\Sigma_{fg}} \vec{v}\cdot(\vec{\bar{\tau}}\cdot\vec{n})\,\partial\Sigma + \int_{V_{fg}} \vec{v}\cdot\rho\vec{f}_m\,\partial V - \int_{\Sigma_{fg}} \vec{q}\cdot\vec{n}\,\partial\Sigma + \int_{V_{fg}} \rho q_{rc}\,\partial V$$

$$\left| \frac{\partial}{\partial t}\left[\rho\left(e + \frac{|\vec{v}|^2}{2}\right)\right] + \nabla\cdot\left(\rho\left(e + \frac{|\vec{v}|^2}{2}\right)\vec{v}\right) = -\nabla\cdot(p\vec{v}) + \nabla\cdot(\vec{\bar{\tau}}\cdot\vec{v}) + \rho\vec{v}\cdot\vec{f}_m - \nabla\cdot\vec{q} + \rho q_{rc} \right| \partial V$$

$V_{fg}(t)$

$$\left(e + \frac{|\vec{v}|^2}{2}\right)\frac{\partial\rho}{\partial t} + \rho\frac{\partial}{\partial t}\left(e + \frac{|\vec{v}|^2}{2}\right) + \rho\vec{v}\cdot\nabla\left(e + \frac{|\vec{v}|^2}{2}\right) + \left(e + \frac{|\vec{v}|^2}{2}\right)\nabla\cdot(\rho\vec{v}) \Rightarrow \boxed{\rho\frac{D}{Dt}\left(e + \frac{|\vec{v}|^2}{2}\right) = -\nabla(p\vec{v}) + \nabla\cdot(\vec{\bar{\tau}}\cdot\vec{v}) + \rho\vec{f}_m\cdot\vec{v} - \nabla\cdot\vec{q} + \rho q_{rc}}$$

0 ; $\rho\frac{D}{Dt}\left(e + \frac{|\vec{v}|^2}{2}\right)$

$$\begin{cases} \text{• Cont.: } \frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\vec{v}) = 0 \xrightarrow{\text{① const. } \rho} \nabla\cdot\vec{v} = 0 \\ \text{• Mom.: } \rho\frac{D\vec{v}}{Dt} = -\nabla p + \nabla\cdot\vec{\bar{\tau}} + \rho\vec{f}_m \quad ② \\ \text{• Energy: } \rho\frac{D}{Dt}\left(e + \frac{|\vec{v}|^2}{2}\right) = -\nabla(p\vec{v}) + \nabla\cdot(\vec{\bar{\tau}}\cdot\vec{v}) + \rho\vec{f}_m\cdot\vec{v} - \nabla\cdot\vec{q} + \rho q_{rc} \quad ③ \end{cases}$$

"$\rho\frac{\partial\vec{v}}{\partial t} + \rho(\vec{v}\cdot\nabla)\vec{v}$" ; $\nabla\cdot[\mu(\nabla\vec{v} + \nabla\vec{v}^T) + \lambda(\nabla\cdot\vec{v})]$

+ 2 equations of state

- liq: $\rho = \text{constant}$, $e = cT + e_0$
- gas: $p = \rho r T$, $e = c_v T + e_0$

$$\begin{cases} \vec{\bar{\tau}} = \mu\left(\nabla\vec{v} + \nabla\vec{v}^T\right) + \lambda\nabla\cdot\vec{v} \quad \text{(Navier-Poisson's Eq)} \\ \vec{q} = -k\nabla T \quad \text{(Fourier's Law)} \end{cases}$$

$\vec{v}\cdot\left\{\rho\frac{D\vec{v}}{Dt} = -\nabla p + \nabla\cdot\vec{\bar{\tau}} + \rho\vec{f}_m\right\}$

$$\left[\rho\frac{D\frac{|\vec{v}|^2}{2}}{Dt} = -\vec{v}\cdot\nabla p + \vec{v}\cdot\nabla\cdot\vec{\bar{\tau}} + \rho\vec{v}\cdot\vec{f}_m\right] ④ \Rightarrow \boxed{\rho\frac{De}{Dt} = -\nabla\cdot(p\vec{v}) + \vec{v}\cdot\nabla p + \nabla\cdot(\vec{\bar{\tau}}\cdot\vec{v}) - \vec{v}\cdot\nabla\cdot\vec{\bar{\tau}} - \nabla\cdot\vec{q} + \rho q_{rc}}$$

$-p(\nabla\cdot\vec{v}) - \vec{v}\cdot\nabla p + \vec{v}\cdot\nabla p$; $\vec{v}\cdot(\nabla\cdot\vec{\bar{\tau}}) + \vec{\bar{\tau}}:\nabla\vec{v} - \vec{v}\cdot(\nabla\cdot\vec{\bar{\tau}})$; Φ_v

$$\begin{cases} \rho\frac{D}{Dt}\left(\frac{|\vec{v}|^2}{2}\right) = -\vec{v}\cdot\nabla p + \vec{v}\cdot(\nabla\cdot\vec{\bar{\tau}}) + \rho\vec{v}\cdot\vec{f}_m \rightsquigarrow \text{Dependent of ② → Mostly not used} \\ \rho\frac{D}{Dt}(e) = -p\nabla\cdot\vec{v} + \vec{\bar{\tau}}:\nabla\vec{v} - \nabla\cdot\vec{q} + \rho q_{rc} \rightsquigarrow \text{Mostly used instead of the complete one} \end{cases}$$

(ωT) ; $\Phi_v > 0$; $(-k\nabla T)$

(constant ρ, μ, c, k)

$$\begin{cases} \nabla\cdot\vec{v} = 0 \\ \rho\frac{D\vec{v}}{Dt} = -\nabla p + \mu\nabla^2\vec{v} + \rho\vec{f}_m \\ \rho c\frac{DT}{Dt} = k\nabla^2 T + \Phi_v + \rho q_{rc} \end{cases}$$

$\vec{f}_m = -\nabla U$ (conservative) ; $\underbrace{-\nabla p - \rho\nabla U}_{-\nabla P} + \mu\nabla^2\vec{v}$

NS 1

- Steady
- Planar
- Perfect liq (const. ρ, μ, k, c)

(as $y \to \infty$, $\begin{cases} V_x = V_\infty \\ P + \rho g z = P_\infty \\ T = T_\infty = T_{wall} \end{cases}$)

$\vec{g}$; $\vec{g} = -g\vec{e}_z$; $\vec{f}_m = \vec{g}$

Cont. $\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} = 0 \rightarrow \boxed{V_z = -V_w}$

$V_x(z \to \infty) = U_\infty$; $V_x(z=0) = 0$

x-mom. $\rho\frac{\partial V_x}{\partial t} + \rho\left(V_x\frac{\partial V_x}{\partial x} + V_y\frac{\partial V_x}{\partial y} + \underbrace{V_z}_{-V_w}\frac{\partial V_x}{\partial z}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 V_x}{\partial x^2} + \frac{\partial^2 V_x}{\partial y^2} + \frac{\partial^2 V_x}{\partial z^2}\right) \Rightarrow -\rho V_w\frac{\partial V_x}{\partial z} = -\frac{\partial p}{\partial x}^{\,0} + \mu\frac{\partial^2 V_x}{\partial z^2} \Rightarrow \frac{\partial^2 V_x}{\partial z^2} - \frac{V_w}{\nu}\frac{\partial V_x}{\partial z} = 0$

$\left[\frac{\partial}{\partial z}\right]\cdot\left[\frac{\partial}{\partial z}\right]\left[\frac{\partial}{\partial z}\right]$; $\left[\frac{\partial}{\partial z^2}\right]\cdot\left[\frac{\partial}{\partial z}\right]\left[\frac{\partial}{\partial z}\right]$; $\left[\frac{\partial}{\partial z^2}\right] = [\nu]$

$\nu = \frac{\mu}{\rho}$

y-mom. $\rho\frac{\partial V_y}{\partial t} + \rho\left(V_x\frac{\partial V_y}{\partial x} + V_y\frac{\partial V_y}{\partial y} + V_z\frac{\partial V_y}{\partial z}\right) = -\frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 V_y}{\partial x^2} + \frac{\partial^2 V_y}{\partial y^2} + \frac{\partial^2 V_y}{\partial z^2}\right) \Rightarrow V_y = 0$

z-mom. $\rho\frac{\partial V_z}{\partial t} + \rho\left(V_x\frac{\partial V_z}{\partial x} + V_y\frac{\partial V_z}{\partial y} + V_z\frac{\partial V_z}{\partial z}\right) = -\frac{\partial p}{\partial z} + \mu\nabla^2 V_z - \rho g \rightarrow 0 = -\frac{\partial p}{\partial z} - \rho g$

$\frac{\partial p}{\partial z} = -\rho g$

$T(z=0) = T_w$ ✓

$v_z(z=0) = -v_w$ ✓

$\bar{v}(z)$?

$T(z)$?

$p(x,z)$?

$p(x,z) = -\rho g z + f(x)$

$p(x,z) + \rho g z = p_\infty \rightarrow \frac{\partial p}{\partial x} = 0$

$$\begin{cases} \frac{\partial^2 v_x}{\partial z^2} + \frac{v_w}{\nu}\frac{\partial v_x}{\partial z} = 0 \\ v_x(z=0) = 0 \\ v_x(z\to\infty) = U_\infty \end{cases} \longrightarrow v_x(z) = C_1 + C_2 \exp\left(-\frac{v_w}{\nu} z\right)$$

$v_x(z=0) = C_1 + C_2 = 0$

$v_x(z\to\infty) = C_1 = U_\infty$

$$\Rightarrow \boxed{v_x(z) = U_\infty\left[1 - \exp\left(-\frac{v_w}{\nu} z\right)\right]}$$

Streamline $\quad \frac{\partial x}{v_x} = \frac{\partial z}{v_z}$

$$\frac{\partial x}{U_\infty\left(1-\exp\left(-\frac{v_w}{\nu} z\right)\right)} = \frac{\partial z}{-v_w} \Rightarrow -v_w \partial x = U_\infty\left(1 - \exp\left(-\frac{v_w}{\nu} z\right)\right)\partial z$$

Heat Flux from fluid to wall $\Rightarrow q_n = \bar{q}\cdot(-\hat{e}_z) = h\Delta T\cdot(-\hat{e}_z) = -k\left.\frac{\partial T}{\partial z}\right|_{z=0}$

energy. $-\rho c v_w \frac{\partial T}{\partial z} = \Phi_v + k\frac{\partial^2 T}{\partial z^2}$

$\Phi_v = \mu\left(\frac{\partial v_x}{\partial z}\right)^2 = \mu\frac{U_\infty^2 v_w^2}{\nu^2}\exp\left(-\frac{2v_w}{\nu} z\right) \quad \left(Pr = \frac{\nu}{\alpha}\right)$

$$\frac{\partial^2 T}{\partial z^2} + \frac{\rho c v_w}{k}\frac{\partial T}{\partial z} = -\frac{\mu U_\infty^2 v_w^2}{k\nu^2}\exp\left(-\frac{2v_w}{\nu} z\right) \rightarrow \begin{cases} \frac{\partial^2 T}{\partial z^2} + \frac{v_w}{\alpha}\frac{\partial T}{\partial z} = -\frac{\mu U_\infty^2 v_w^2}{k\nu^2}\exp\left(-\frac{2v_w}{\nu} z\right) \\ T(z=0) = T_w \\ T(z\to\infty) = T_\infty \end{cases}$$

$T(z) = C_1' + C_2'\exp\left(-\frac{v_w}{\alpha} z\right) + T_p$

$T_p = C_3\exp\left(\frac{2v_w}{\nu} z\right)$

$$\frac{T(z)-T_\infty}{T_w - T_\infty} = 1 + \frac{U_\infty^2}{c(T_w - T_\infty)}\frac{Pr}{2(2-Pr)}\exp\left(-\frac{v_w z}{\alpha}\right) - \frac{U_\infty^2}{c(T_w-T_\infty)}\frac{Pr}{2(2-Pr)}\exp\left(\frac{2v_w}{\nu} z\right)$$

NS 3

$|\bar{x}| \to \infty$

$\bar{v} = 0$ $\quad v_r(r\to\infty, t) \to 0$

$p(r\to\infty, t) \to p_\infty$

$\bar{v} = v_r(r,t)\hat{e}_r$



$v_r(r,t)$?

$p(r,t)$?

$\dot{W}_{in fluid}$?

fluid ρ, μ (constant)

- Spherical sym. $\frac{\partial}{\partial\theta} = \frac{\partial}{\partial\varphi} = 0 \rightarrow v_\theta = v_\varphi = 0$
- Buoyancy force $\rightarrow$ neglect $\bar{g}$

$\left(v_r(r=R,t) = \dot{R} = \frac{\partial R}{\partial t}\right)$

cont. $\frac{\partial}{\partial r}(r^2 v_r) = 0 \Rightarrow r^2 v_r = C(t) \Rightarrow R^2\dot{R} = C(t) \Rightarrow \boxed{v_r(r,t) = \frac{R^2}{r^2}\dot{R}}$

$(r \geq R(t))$

r- moment. $\rho\frac{\partial v_r}{\partial t} + \rho v_r\frac{\partial v_r}{\partial r} = -\frac{\partial p}{\partial r} \Rightarrow \frac{\partial p}{\partial r} = \rho\left(\frac{2R\dot{R}^2}{r^2} + \frac{R^2}{r^2}\ddot{R}\right) - \rho\frac{R^4\dot{R}^2}{r^5}\frac{(-2)R^2 R}{r^3} \Rightarrow \boxed{p(r,t) = p_\infty = \rho\left(\frac{2R\dot{R}^2 + R^2\ddot{R}}{r} - \frac{R^4\dot{R}^2}{2r^4}\right)}$

$\dot{W}_{balloon \to liq} = -\int_{S_b}\bar{v}\cdot p\bar{n}\,dS + \int_{S_b}\bar{\bar{\tau}}'\cdot\bar{n}\cdot\bar{v}\,dS = +v_r(r=R,t)\,p(r=R,t)\,4\pi R^2 + \frac{4\mu\dot{R}}{R}4\pi R^2 \Rightarrow \dot{W}_{liq} = \left(p_\infty\dot{R} + \frac{3}{2}\rho\dot{R}^3 + \rho R\dot{R}\ddot{R} + \frac{4\mu\dot{R}^3}{R}\right)4\pi R^2$

$(\bar{n} = -\hat{e}_r)$

$\otimes_1$

$\otimes_1 \quad \left.\bar{\bar{\tau}}'\right|_{r=R} = \begin{pmatrix} 2\mu\frac{\partial v_r}{\partial r} & 0 & 0 \\ 0 & 2\mu\frac{v_r}{r} & 0 \\ 0 & 0 & 2\mu\frac{v_r}{r} \end{pmatrix}_{r=R} = 2\mu\left.\begin{pmatrix} -2\frac{R^2\dot{R}}{r^3} & 0 & 0 \\ 0 & \frac{R^2\dot{R}}{r^3} & 0 \\ 0 & 0 & \frac{R^2\dot{R}}{r^3} \end{pmatrix}\right|_{r=R} = 2\mu\frac{\dot{R}}{R}\begin{pmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow \bar{\bar{\tau}}'\cdot\bar{n} = 2\mu\frac{\dot{R}}{R}\begin{pmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} = \frac{4\mu\dot{R}}{R}\hat{e}_r \Rightarrow (\bar{\bar{\tau}}'\cdot\bar{n})\cdot\bar{v} = \frac{4\mu\dot{R}^2}{R}$